DSA Prep

Interview OS — a short user guide

Offline-capable, installable, no account. Everything runs in your browser.

486	5	42	46
problems	nested phases	company sheets	animated algorithms

Running it

From the project folder:

`npm install`	Once, after cloning.
`npm run dev`	Development server on `http://localhost:3000`.
`npm run build`	Production build. Prerenders every page.
`npm start`	Serves the production build. Offline mode works here.

Needs Node 20 or newer. There is no backend, no database and no environment variables to set.

> **Install it as an app.** On a computer, use the install button in the Chrome, Edge or Brave address bar. On an iPhone, tap Share then *Add to Home Screen* in Safari; on Android, Chrome's menu has *Install app*. It then opens in its own window and works with no network. Install from the same address you practise on: progress belongs to that address, and Profile → Your data shows which one it is.

First run

A five-step wizard appears the first time. It is worth two minutes, because the answers seed the learning engine.

1. **Your name** — only used to greet you, and optional.
2. **Your level** — sets your starting rating, so the first recommendations are not random.
3. **Target company** — reweights every topic towards that company's interview style. Skippable.
4. **Runway** — how long until your interview. This picks your starting phase.
5. **Daily minutes** — the planner sizes each day to exactly this budget.

All five are editable later from **Profile** and from the dashboard.

The daily loop

Open the **Dashboard** and work top to bottom. It is designed so you never have to decide what to do next.

DAILY CHALLENGE

One problem, date-seeded

The same for everyone, rotating by weekday: easy on Sunday, hard on Wednesday and Saturday, medium otherwise. Solve it, then claim the bonus XP.

TODAY'S PLAN

Reviews first, then new work

Due reviews come first, the remaining minutes are filled with the highest-value problems that fit, and one deliberate stretch problem is added at the end.

GOAL RING

Problems solved today

Counts towards your streak. Streaks carry two freezes, and each absorbs one missed day.

RECOMMENDED NEXT

Scored, not shuffled

Each card says why it was picked: pattern mastery, your rating, interview frequency, your target company.

Phases: 75 → 150 → 250 → 450 → Vault

The sheets are strictly nested, so moving up a phase never asks you to redo anything.



Every problem in Blind 75 is also in Top 150, which sits inside SDE 250, inside DSA 450, inside the Vault. A problem you solved in an earlier phase shows a **done in 75** badge in every larger sheet. Switch your active phase from the dashboard or from the **Sheets** page.

Company sheets

Pick a company from **Companies** and you get its real round structure, what its bar looks like, its difficulty mix, a reordered problem list, and a readiness score out of 100 that compares your mastery and rating against that company's tier. Service-company pages (TCS, Infosys, Wipro and similar) are tuned for online assessment rounds rather than full onsite loops.

Set one as your **target** and the whole app reorders around it: recommendations, the daily plan and the weakest-skills list all shift towards that company's focus topics.

Progress, XP and streaks

Solving awards XP with a visible breakdown: difficulty, a no-hint bonus, a first-try bonus, your streak multiplier, and a speed bonus for finishing inside the estimate. XP drives 24 levels, each with its own title, and there are 42 achievements across four tiers — three of them secret.

Behind all of it, four models run on your history: a spaced-repetition scheduler, Bayesian knowledge tracing per topic and per pattern, an Elo rating for you against every problem, and a planner that fills each day with a knapsack over your remaining minutes. None of them need a server.

Working a problem

Every problem page has the same five parts.

Interview mode	A timed round scaled to the problem's target time, with phase prompts: clarify, plan out loud, code, verify, wrap up. Finish as *solved* or *attempted* and the time is recorded either way.
Hint ladder	Hints reveal one at a time, from a nudge to the full approach. Using fewer hints earns more XP and counts as stronger evidence of mastery.
Quick refresher	Every problem has a one-minute recap: the key idea, the steps, the usual mistake and the complexity. It opens by itself once you have solved the problem, and the note icon on any row shows it in a pop-up.
Approach	The intended solution, its complexity, and a follow-up an interviewer would ask next.
Scratchpad	A code editor with syntax highlighting. JavaScript actually runs, in a sandboxed worker with a timeout.
Notes	Saved as you type. These collect in the Notebook and are the best revision material in the app.

> **Find the same problem elsewhere.** Each page links out to LeetCode, GeeksforGeeks, Code360, InterviewBit, HackerRank, CodeChef and a video search. Links are marked *direct* when the exact problem is known and *search* otherwise, so you always know which is which.

> **Getting back.** Every detail page has a **Back** button, and next to it a named chip linking straight to the list that sent you there. The breadcrumb beside them is a category path, so the middle crumb on a problem takes you to that topic in Learn, which is deliberate rather than a way back.

Spaced repetition

Solving a problem schedules it for review. The scheduler tracks a memory *stability* per problem and brings it back when your predicted recall falls to about 90 percent — the point where one look is enough and any later would mean relearning it from scratch.

Open **Revise** and grade each card honestly: **Forgot**, **Hard**, **Good** or **Easy**. Grading low is not a failure, it just pulls the card closer. Reviews count towards your streak.

Seeing the algorithms

Forty-six step-by-step animations, all with editable input.

VISUALISE

One algorithm, frame by frame

Play, pause, step, scrub and replay at seven speeds. Each frame narrates what just happened, highlights the pseudocode line being executed, and updates live counters. Type your own input and watch it run on that.

ALGORITHM RACE

Two algorithms, same input

Pick a matchup — bubble against quick, BFS against DFS — and step both in lockstep. The verdict says how many times fewer steps the winner took. Steps are animation frames, so they track comparisons and visits rather than nanoseconds.

The other tools

Roadmap	The 24 topics as a prerequisite graph, seven layers deep. Nodes fill as your mastery grows, and a lock means a prerequisite is not ready. Hover any topic to trace what it needs and what it unlocks.
Complexity explorer	Drag *n* and watch which growth classes fall off the cliff. The constraint decoder turns `n ≤ 10^5` into a shortlist of algorithms that will actually pass — the single most useful reflex in a live round.
Notebook	Every note you have written, everything you starred, every problem you attempted but did not finish, and your recent history. Searchable, editable in place, and exportable as one Markdown file.
Arena	Three games: a timed mock round drawn from a company's real difficulty mix, a pattern-recognition sprint, and a complexity reading drill.
Focus timer	The pill in the bottom-left corner. Sessions of 25 or 50 minutes with breaks, and it keeps running while you browse. Completed sessions log study minutes towards your streak.
Learn	Notes per topic and per pattern: the key ideas, the trigger phrases that give the pattern away, a code template and a generic hint ladder.
Analytics	Mastery radar, rating history, an activity heatmap and difficulty breakdowns.

Keyboard shortcuts

Press `?` anywhere to see this list inside the app.

Jump to a page

`g` `d`	Dashboard	`g` `r`	Revise
`g` `s`	Sheets	`g` `a`	Arena
`g` `c`	Companies	`g` `n`	Notebook
`g` `p`	Problems	`g` `m`	Roadmap
`g` `v`	Visualise	`g` `y`	Analytics
`g` `l`	Learn	`g` `u`	Profile

Anywhere

`Ctrl` `K`	Search everything
`/`	Search everything
`f`	Focus timer
`t`	Next theme
`?`	Shortcut cheat sheet
`Space`	Play a visualiser
`←` `→`	Step one frame

The search box also runs actions: pick a theme, start the focus timer, race two algorithms, or jump to a random problem.

Your data

Everything — progress, notes, code, settings, streak — lives in this browser's local storage under one key, `dsa-prep-v1`. Nothing is uploaded anywhere, and there is no account to create.

- **Moving devices:** Profile → Export downloads a JSON file. Import it on the other device.
- **Same browser:** your streak only counts if you use the same browser profile each day.
- **Starting over:** Profile → Reset clears progress. It cannot be undone, so export first.
- **Private windows:** storage is cleared when the window closes, so progress will not persist.
- **Done on LeetCode:** the button on every row, or a pasted list in Profile, marks what you have already beaten. It is kept apart from the sheet tick, and those problems are swapped for similar ones.

A suggested first week

1. Finish the wizard. Pick the runway that is actually true, not the one you wish were true.
2. Do the daily challenge, then whatever the plan lists. Do not skip ahead — the order is doing work.
3. Write one line of notes on every problem, even the easy ones. Future you reads those, not the solution.
4. When a problem confuses you, open its visualiser before you read the approach.
5. Clear the review queue daily. It is short by design, and skipping it is what makes phase two feel hard.
6. On day seven, open the Roadmap and Analytics. Both will show you what to fix without you having to guess.

> **One honest caveat.** Problem metadata is curated, not scraped. Interview frequency is a judgement call on a five-point scale, company tags are reported tags where those are known, and round structures reflect publicly reported processes that drift over time. Treat the numbers as a good prior, not as ground truth.

If something looks wrong

Empty dashboard	Nothing is solved yet. The plan fills in as soon as you mark the first problem.
Streak reset	Check you are in the same browser profile. Freezes cover one missed day each, not two.
App opens empty	It was installed from a different address. Profile → Your data shows the one in use; reinstall from the address that has your progress.
Wizard reappears	Site data was cleared. Import your exported JSON from Profile.
No offline mode	Offline needs the production build: `npm run build` then `npm start`.

Making it yours

Seven themes, in **Profile** → **Settings**, the **Theme** picker at the top of the sidebar, or `t` to cycle. Each one sets the sidebar, the page, the accent and the landscape on the dashboard. Difficulty colours stay the same in all of them, because they carry meaning.

